

Process Hierarchies

Definition

Process hierarchy describes the parent-child relationships formed when a process creates another process. A child process can create more processes, producing a tree-like structure of descendants.

1. Basic Idea

Main Concept

In some operating systems, a process remains associated with the process it created. This creates a **parent-child relationship**, and repeated process creation forms a **hierarchy**.

Key Property

Each process has only **one parent**, but it may have:

- no children,
- one child,
- or many children.

Tree Structure

Because each process has a single parent, process hierarchies are organized like a **tree**, not a network. One process sits above its descendants, and children may create further descendants of their own.

2. UNIX Process Hierarchies

UNIX View

In UNIX, a process and all of its children and later descendants form a structured hierarchy. Related processes are often treated together as part of a **process group**.

Process Group

A **process group** in UNIX is a set of related processes that can receive signals together. This is especially important for command-line sessions.

Keyboard Signals

When a user sends a signal from the keyboard, the signal is usually delivered to all processes in the process group currently attached to that terminal or window.

How a Process Responds

After receiving a signal, each individual process may:

- catch the signal and handle it,
- ignore the signal,
- or take the default action, which is often termination.

3. UNIX System Startup Example

How UNIX Builds the Process Tree

After the computer boots, UNIX starts a special process called **init**. This process helps build the system's overall process hierarchy.

Startup Sequence

The basic sequence is:

1. **init** starts running from the boot image.
2. It reads configuration information, such as how many terminals exist.
3. It creates one process for each terminal.
4. Each of those processes waits for a user to log in.
5. After a successful login, the login process starts a shell.
6. The shell may create more processes when the user runs commands.

Result

Because each process is created by an earlier one, all processes in the UNIX system can be viewed as part of one large tree, with **init** at the root.

4. Windows and Process Hierarchies

Important Difference

Windows does **not** have a true process hierarchy in the UNIX sense. All processes are treated as fundamentally equal.

Parent Handle

When a Windows process creates another process, the parent receives a special token called a **handle**. This handle can be used to control the child process.

Why This Is Not a Real Hierarchy

The parent can pass that handle to another process. Once that happens, the original parent-child control relationship is no longer strict, so the hierarchy is not preserved in the same way as in UNIX.

UNIX vs Windows

In UNIX, a process cannot simply disinherit its children. In Windows, control can effectively be transferred through handles, so the parent-child relationship is weaker.

5. UNIX vs Windows Summary

System	Hierarchy Model	Key Idea
UNIX	True process tree	Parent-child relationships remain meaningful, and all system processes can be viewed as descendants in one tree.
Windows	No strict hierarchy	Processes are equal; control over a child depends on handles, which can be passed to other processes.

6. Why Process Hierarchies Matter

Practical Importance

Process hierarchies help the operating system:

- organize related processes,
- manage groups of processes together,
- deliver signals to a whole process group,
- and describe system startup in a clear tree structure.

7. Quick Review

Exam-Focused Points

- A process hierarchy forms when parent and child processes remain associated.
- A process has only one parent, but it may have many children.
- In UNIX, processes and their descendants form a meaningful tree structure.
- UNIX process groups allow keyboard signals to be delivered to related processes together.
- Each UNIX process may catch, ignore, or take the default action for a signal.
- During UNIX startup, **init** acts as the root of the system's process tree.
- Windows does not maintain a strict process hierarchy like UNIX.
- In Windows, handles can be passed to other processes, weakening the original parent-child relationship.